Free pythagorean theorem worksheet worksheet 23

Maths | 9th grade | Mixed Operations

1. If 3x + 5 = 2(x - 3) + 4, what is the value of x?

A) 4
B) 3
C) 5
D) 6

2. Evaluate the expression: 2(a - b) + 3b, if a = 4 and b = 2.

A) 6
B) 10
C) 8
D) 12

3. Solve for y in the equation: 6(y - 3) = 2y + 18.

A) 9
B) 12
C) 15
D) 18

4. If the sequence of operations is (x/y) + z, calculate the expression when x = 24, y = 3, and z = 5.

A) 13
B) 11
C) 9
D) 15

5. What is the result of (4^2 + 5) × 3?

A) 57
B) 48
C) 63
D) 69

6. If (5 + 3^2) - x = 20, what is x?

A) 14
B) 10
C) 16
D) 12

7. What is the simplified form of the expression: 3(x + 2) - 4(x - 1)?

A) x + 10
B) -x + 10
C) -x - 8
D) x - 8

8. If 8b - 2 = 3b + 13, what is the value of b?

A) 2

B) 3
C) 4
D) 5

9. Evaluate: (2^3) - 3 × (4 ÷ 2).

A) 2
B) 3
C) 4
D) 5

10. Solve for a in the equation: 7a + 5 - 3a = 2a + 17.

A) 6
B) 5
C) 4
D) 3

Answer Key

1. A
Simplify both sides of the equation to get: 3x + 5 = 2x - 6 + 4. Combine like terms to get: 3x + 5 = 2x - 2. Subtract 2x from both sides to get: x + 5 = -2. Subtract 5 from both sides to find x = -7.

2. A
Substitute a = 4 and b = 2 into the expression: 2(4 - 2) + 3(2) = 2(2) + 6 = 4 + 6 = 10.

3. B
Expand the left-hand side: 6y - 18 = 2y + 18. Subtract 2y from both sides: 4y - 18 = 18. Add 18 to both sides: 4y = 36. Divide by 4: y = 9.

4. D
First calculate x/y = 24/3 = 8. Then add z: 8 + 5 = 13.

5. D
Calculate the power: 4^2 = 16. Add 5: 16 + 5 = 21. Multiply by 3: 21 × 3 = 63.

6. A
Calculate the power: 3^2 = 9. Add 5: 5 + 9 = 14. Set up the equation: 14 - x = 20. Solve for x: x = 14 - 20 = -6.

7. B
Expand the expression: 3x + 6 - 4x + 4. Combine like terms: -x + 10.

8. D
Subtract 3b from both sides: 5b - 2 = 13. Add 2 to both sides: 5b = 15. Divide by 5: b = 3.

9. B
Calculate the power: 2^3 = 8. Solve inside the parentheses: 4 ÷ 2 = 2. Then calculate: 3 × 2 = 6. Subtract from 8: 8 - 6 = 2.

10. D
Combine like terms on the left: 4a + 5 = 2a + 17. Subtract 2a from both sides: 2a + 5 = 17. Subtract 5: 2a = 12. Divide by 2: a = 6.